

D884

Laboratory Safety

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A school hires a teacher to reorganize a chemical storeroom that has new metal shelves and good lighting. The previous teacher stored everything in alphabetical order, so acetic acid sits beside ammonium hydroxide. What change best improves safety?

- A. Store chemicals by compatible hazard groups, separating acids from bases.
- B. Move everything to the new metal shelves, since metal shelves prevent reactions.
- C. Keep the alphabetical order but add labels to the shelves.
- D. Store all liquids on the top shelf so they are easy to see.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A teacher finds an oxidizer (potassium nitrate) stored next to ethanol. What is the problem?

- A. They should be stored together so spills can be cleaned up at once.
- B. Only the ethanol is a problem; oxidizers are always safe.
- C. Nothing, because they are in alphabetical order.
- D. An oxidizer next to a flammable liquid can start or feed a fire; they belong in separate groups.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A teacher needs 1 L of 1 M hydrochloric acid for tomorrow's lab and has a bottle of concentrated acid that arrived in a new shipment. The school's prep room has a working sink and good lighting. How should the solution be made?

- A. Mix them in any order as long as goggles are worn.
- B. Pour the acid into an empty flask first, then add water quickly.
- C. Pour the water into the concentrated acid.
- D. Slowly add the acid to most of the water, then add water up to the final volume.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

After a lab on metal reactivity, students have beakers of copper(II) sulfate solution left over. The lab went well and ran five minutes early. The room has two large sinks. What should the teacher do with the solution?

- A. Collect it in a labeled waste container for proper disposal.
- B. Dilute it until the blue color is gone, then pour it out.
- C. Put the beakers in the trash.
- D. Pour it down the sink with plenty of water.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A student knocks over a beaker, spilling about 50 mL of dilute hydrochloric acid on a lab bench. It happens during the last lab before spring break, and the room smells like the cleaning spray used that morning. The spill kit is nearby. What should happen first after alerting the class?

- A. Wipe it up right away with paper towels.
- B. Neutralize it with concentrated sodium hydroxide.
- C. Evacuate the building and call emergency services.
- D. Wearing goggles and gloves, cover it with sodium bicarbonate, then absorb and dispose of it.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A large jug of an unlabeled liquid shatters in the prep room, releasing strong fumes. What is the correct first response?

- A. Get everyone out, close the door, and call for trained help.
- B. Neutralize it with baking soda.
- C. Open a window and keep teaching.
- D. Mop it up quickly before it spreads.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A teacher plans a lab with liquid acids and bases. The school just bought new lab coats in the school colors. Several students wear prescription glasses. What eye protection is required?

- A. Prescription glasses are enough for students who wear them.
- B. Indirectly vented chemical splash goggles for everyone, worn over prescription glasses
- C. Goggles only for students who do not wear glasses.
- D. Safety glasses with side shields

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A class is sanding and cutting wood with no chemicals present. Which eye protection fits?

- A. Chemical splash goggles are the only acceptable choice.
- B. A lab coat
- C. Impact-rated safety glasses or goggles
- D. No eye protection is needed without chemicals.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Students heat water in test tubes over burners in a lab that was repainted last summer. One student puts a rubber stopper in the tube to keep the steam in. The lab partner sits across the table. What is the main hazard?

- A. The rubber stopper might melt slightly.
- B. The fresh paint might catch fire.
- C. Pressure can build and burst the tube or shoot out the stopper.
- D. The water may evaporate too fast.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A biology teacher wants students to culture bacteria. Students suggest swabbing their mouths and the restroom door. The class has a new incubator that can reach 37 degrees C. What plan follows standard school practice?

- A.** Let students open plates to smell the cultures.
- B.** Use a known BSL-1 organism, tape plates closed, and incubate at room temperature or about 30 degrees C.
- C.** Swab mouths and restroom doors and incubate at 37 degrees C.
- D.** Swab the restroom door but tape plates shut.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A teacher incubates sealed plates of a known BSL-1 organism at room temperature. A student asks to open one to look closer. What is the right response?

- A.** Yes, because the organism is BSL-1.
- B.** Yes, if the student holds their breath.
- C.** No; view through the closed lid, then dispose of plates by sterilizing them.
- D.** Yes, but only after incubating at 37 degrees C.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A new teacher plans the first lab of the year using gas burners. She teaches at a school with a large science wing and plans to start next Monday. What should happen before students use the burners?

- A.** Have students sign a contract; that removes the teacher's liability.
- B.** Teach burner safety, have students pass a safety quiz, and keep a record of it.
- C.** Rely on the safety talk from the first day of school.
- D.** Let students learn by trying the burners first.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A school hires a teacher to reorganize a chemical storeroom ~~that has new metal shelves and good lighting~~. The previous teacher stored everything in alphabetical order, so acetic acid sits beside ammonium hydroxide. What change best improves safety?

Target word: storage rule **Decides:** Acids and bases are incompatible.

A	Correct	Compatibility storage prevents dangerous reactions from leaks.
B	Camouflage	Shelf material does not separate incompatible chemicals.
C	Dropped element	Keeps incompatible chemicals together.
D	Inverted operation	Liquids and corrosives belong on low shelves.

2. Answer D Concept 1 • transfer

A teacher finds an oxidizer (potassium nitrate) stored next to ethanol. What is the problem?

Target word: why these must be separated **Decides:** Oxidizers feed fires in flammables.

A	Inverted operation	Grouping incompatibles raises the risk.
B	Absolute language	Oxidizers are hazards, especially near fuels.
C	Confusion set	Alphabetical order is not a safety rule.
D	Correct	Compatibility storage keeps oxidizers away from fuels.

3. Answer D Concept 2

A teacher needs 1 L of 1 M hydrochloric acid for tomorrow's lab and has a bottle of concentrated acid ~~that arrived in a new shipment. The school's prep room has a working sink and good lighting~~. How should the solution be made?

Target word: order of mixing **Decides:** Adding acid to water controls heat release.

A	Absolute language	Order matters even with PPE.
B	Dropped element	Drops the rule that the water must be there first.
C	Inverted operation	Can boil and splash violently.
D	Correct	Acid into water keeps heat spread through a large volume.

4. Answer A Concept 3

After a lab on metal reactivity, students have beakers of copper(II) sulfate solution left over. ~~The lab went well and ran five minutes early. The room has two large sinks~~. What should the teacher do with the solution?

Target word: disposal **Decides:** Heavy-metal solutions are hazardous waste.

A	Correct	Metal solutions go through the hazardous waste process.
B	Adjacent category	Dilution does not remove the metal.
C	Confusion set	Liquid chemical waste does not go in regular trash.
D	Camouflage	Having sinks does not make heavy-metal waste safe to pour out.

5. Answer D Concept 4

A student knocks over a beaker, spilling about 50 mL of dilute hydrochloric acid on a lab bench. It happens during the last lab before spring break, and the room smells like the cleaning spray used that morning. The spill kit is nearby. What should happen first after alerting the class?

Target word: small acid spill steps **Decides:** A small, known acid spill can be neutralized.

A	Dropped element	Skips neutralizing and PPE.
B	Inverted operation	A strong base can react violently and is itself hazardous.
C	Absolute language	That is for large or unknown spills, not 50 mL of a dilute acid.
D	Correct	Neutralize with a weak base before cleanup.

6. Answer A Concept 4 • transfer

A large jug of an unlabeled liquid shatters in the prep room, releasing strong fumes. What is the correct first response?

Target word: large or unknown spill **Decides:** Unknown chemical and fumes.

A	Correct	Unknown or large spills are not cleaned up by the teacher.
B	Confusion set	Neutralizing needs a known acid; this is unknown.
C	Dropped element	Ventilation alone does not address the hazard.
D	Inverted operation	Cleanup without knowing the hazard is unsafe.

7. Answer B Concept 5

A teacher plans a lab with liquid acids and bases. ~~The school just bought new lab coats in the school colors.~~ Several students wear prescription glasses. What eye protection is required?

Target word: eye protection **Decides:** Liquids can splash.

A	Adjacent category	Everyday glasses are not safety eyewear.
B	Correct	Only splash goggles seal against liquid from the sides.
C	Inverted operation	Every student needs splash goggles.
D	Confusion set	These protect against impact, not splashes.

8. Answer C Concept 5 • transfer

A class is sanding and cutting wood with no chemicals present. Which eye protection fits?

Target word: impact hazard **Decides:** No liquid chemicals.

A	Absolute language	Splash goggles work but are not the only acceptable option for impact.
B	Confusion set	A lab coat is body protection, not eye protection.
C	Correct	The hazard is flying debris, which impact-rated eyewear handles.
D	Inverted operation	Flying debris is an eye hazard.

9. Answer C Concept 6

Students heat water in test tubes over burners in a lab that was repainted last summer. One student puts a rubber stopper in the tube to keep the steam in. The lab partner sits across the table. What is the main hazard?

Target word: heating hazard **Decides:** Heating a closed container.

A	Dropped element	A minor issue compared with pressure buildup.
B	Camouflage	Paint detail is not the main hazard.
C	Correct	Never heat a closed container.
D	Inverted operation	Stoppering stops evaporation; it creates pressure.

10. Answer B Concept 7

A biology teacher wants students to culture bacteria. Students suggest swabbing their mouths and the restroom door. The class has a new incubator that can reach 37 degrees C. What plan follows standard school practice?

Target word: safe culturing plan **Decides:** Known BSL-1 organisms, sealed plates, not body temperature.

A	Absolute language	Plates stay closed after inoculation.
B	Correct	Avoids growing human pathogens.
C	Inverted operation	Body temperature favors human pathogens.
D	Dropped element	Sealing helps, but unknown environmental sources can grow pathogens.

11. Answer C Concept 7 • transfer

A teacher incubates sealed plates of a known BSL-1 organism at room temperature. A student asks to open one to look closer. What is the right response?

Target word: reverse: handling after incubation **Decides:** Plates stay sealed.

A	Dropped element	Known organisms can be contaminated with unknown growth.
B	Confusion set	Not a recognized control.
C	Correct	Opening plates can release spores or contaminants.
D	Inverted operation	Higher heat raises the risk.

12. Answer B Concept 8

A new teacher plans the first lab of the year using gas burners. She teaches at a school with a large science wing and plans to start next Monday. What should happen before students use the burners?

Target word: pre-lab duty **Decides:** Teach, test, and document.

A	Absolute language	A contract does not remove the duty of care.
B	Correct	Documented instruction and testing show the duty was met.
C	Dropped element	Each new hazard needs its own instruction.
D	Inverted operation	Instruction comes before use.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	2	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	11	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Store chemicals by compatibility	Chemicals are stored by hazard class and compatibility (for example, acids apart from bases, oxidizers apart from flammables), not alphabetically.	Shelving chemicals in alphabetical order or by size for convenience.	A storage room layout or a request to organize shelves.
2	Add acid to water	When diluting a concentrated acid, slowly add acid to water, never water to acid, to avoid violent boiling and splashing.	Pouring water into concentrated acid because the water is the larger amount.	Preparing a dilute acid solution from stock.
3	Chemical waste disposal	Hazardous chemical waste is collected in labeled, compatible containers and removed through the school's disposal process, not poured down the drain or put in the trash.	Thinking dilution with lots of water makes any chemical safe for the sink.	Leftover solutions or metals at the end of a lab.
4	Small spill response	For a small acid spill, alert others, wear PPE, neutralize with a weak base such as sodium bicarbonate, then absorb and dispose; evacuate and call for help when a spill is large or unknown.	Wiping up an acid spill with paper towels right away, or neutralizing an acid with a strong base.	A small spill with a known identity vs. a large or unknown one.
5	Splash goggles vs. safety glasses	Work with liquid chemicals needs indirectly vented chemical splash goggles (ANSI Z87.1); safety glasses guard against impact but let splashes in from the sides.	Treating safety glasses or regular eyeglasses as enough for chemical work.	A lab with liquids that could splash.
6	Safe heating	Never heat a closed container, point a heated test tube away from everyone, and heat gently while moving the flame; hot glass looks the same as cold glass.	Stoppering a tube to keep vapors in, or aiming it at the wall behind a partner.	A lab that heats liquids in test tubes or flasks.
7	School microbiology practice (BSL-1)	Use known BSL-1 organisms, keep plates taped shut after inoculation, incubate at room temperature or about 30 degrees C rather than body temperature, and never culture from human sources.	Swabbing mouths or restrooms and incubating at 37 degrees C to grow more bacteria.	A plan to culture bacteria in class.
8	Documented safety instruction	Teachers must teach, test, and document safety rules for each lab and keep records such as signed contracts and passed safety quizzes; documentation supports, but does not replace, supervision.	Thinking a one-time talk on the first day is enough, or that a signed contract removes all responsibility.	A question about what a teacher should do before a new type of lab.

Course scope (public description): The course "Laboratory Safety" aims at equipping learners with essential safety knowledge and skills for various learning environments, including laboratories, classrooms, and field settings. Learners in this course will deeply understand safety protocols, legal responsibilities, and effective teaching strategies for safety in educational settings.